SETAC Theme 1 Master Tables: Metadata

About this document

The Theme 1 workbook (SETAC_Theme1_master_tables.xlsx) has 8 sheets built from 4 original SETAC files, linked by Contact ID. Three sheets are the master tables for analysis, two are the building blocks behind them, one holds the records removed for having no Contact ID, and two are summary counts.

Sheet	One row is	Rows	To be Used for
A_renewal_opportunities	one renewal decision	15,170	Renewal, expiry and reinstatement (Analysis 3), life cycle (5), retention by group (6)
B_year_end_members	one member on one year-end date	16,361	Membership growth (Analysis 1) and composition (2)
C_member_cohorts	one person	11,141	Cohort retention (Analysis 4)
person_details	one contact, member or not	28,922	Building block: personal details added to A, B and C
valid_subscriptions	one membership term	16,046	Building block: the cleaned subscriptions behind A, B and C
excluded_subs_blank_co_id	one subscription record with no Contact ID	2,238	Showing what was removed and checking it for bias.
exclusion_counts	one reason a record was kept or removed	7	Showing what was removed and why
year_end_comparison	one year	4	Showing the effect of removing rows with no Contact ID

The original files

Original file	What one row is	What we took from it
Subscriptions Europe.xlsx	one SETAC Europe subscription	The membership history: class, grade, plan, dates, price
Individual Contacts.xlsx	one contact (includes non-members, staff and suppliers)	Country, sector, graduation date, retired, mentor, topics
Country list.xlsx	one country	SETAC region and fee band

Original file	What one row is	What we took from it
Subscriptions All.xlsx	one subscription in any SETAC region	Only used to spot members who moved to another region

Key terms used in every sheet

Term	Meaning
Subscription record	One row in SETAC's subscription file. A person can have several.
Term	One membership period of 1 or 3 years. SETAC renews a membership inside the same record by moving the renewal date forward, so one record can hold several terms. We split each record into one row per term.
Renewal decision (due date)	The date a term ends. The member either renews or leaves.
Grace period	1 month (31 days), confirmed by SETAC. A member who starts a new term within it counts as renewed.
Lapsed (SETAC's meaning)	The 1-month window after a membership ends, while the member can still renew.
Expired (our outcome)	Did not renew within the 1-month grace period. SETAC treats the membership as expired, and the person must reapply as a new member.
Reinstated	Expired, then came back later with a new membership.
Spell	An unbroken period of membership. A gap of more than 1 month ends a spell; coming back starts a new one.
Merged profile	A duplicate contact profile that SETAC folded into another profile of the same person. Its subscriptions stay under the old ID, which cannot be linked to the surviving one, so they are excluded.
Cut-off	30 September 2026, the last date covered by the data.
Migration	SETAC moved to its current system in June 2023 (contacts created 23 June, subscriptions 26 June). Records without dates in the old system were given an artificial start date of 2 July 2023 and a first renewal date of 1 July 2024.
Legacy products	Explorer and 3-year memberships. SETAC stopped renewing them after 31 August 2026.

How to read the column tables

- **Original** means the column was copied from a SETAC file. The file and its original column name are given.
- **Created** means the column did not exist before. It was calculated in the notebook, and the explanation says how.
- Yes/No columns hold True or False. Flag columns hold 1 (yes) or 0 (no). Money is in euros.

Sheet 1: A_renewal_opportunities

One row is one renewal decision: the end of one membership term, when the member either renewed or left. The sheet has 15,170 rows, 33 columns and 11,137 people. It answers whether members renew, expire or come back later.

Built from

Original file	Columns taken	Joined on
Subscriptions Europe	Contact - ID, ID, Class, Grade, Duration, Subscription plan, Start date, Renewal date, Cancelled at, Current price, Sub status (plus Contact type, Contact status, Status and Date created, used only to filter and flag rows: Contact type removes staff and other non-members, Contact status removes merged and deceased profiles)	The starting table
Individual Contacts	Default address - Country, Sector, Retired or currently not working, Graduation date	Contact ID
Country list	Geo unit, Income level	Country name
Subscriptions All	Subscription plan, Start date, Status	Contact ID, only for people who did not renew

Columns

Column	What it means	Source
subscription_id	Unique ID for this term: the original subscription ID plus "_t" and the term number. "..._t2" is the second term in that record.	Created, from Subscriptions Europe › ID
contact_id	Anonymised ID of the person. Links every sheet together.	Original: Subscriptions Europe › Contact - ID
spell_no	Which spell of membership this is: 1 = first	Created

Column	What it means	Source
	unbroken period, 2 = came back after expiring, up to 4.	
renewal_number	Which renewal decision this is within the spell: 1st, 2nd, 3rd and so on.	Created
member_class	Membership class: Free, Full or Explorer.	Original: Subscriptions Europe › Class
paid_or_free	"Free" for Free members, "Paid" for Full and Explorer.	Created, from Class
grade_short	Grade without the word "Membership": Professional, Student, Recent Graduate, Senior Active, Emeritus.	Created, from Subscriptions Europe › Grade
member_type	Career group for comparisons. Same as grade_short, except people who ticked "Retired or currently not working" are shown as Retired.	Created, from Grade + Individual Contacts › Retired
plan	Membership plan: annual or 3-year, at standard or discounted fee.	Original: Subscriptions Europe › Subscription plan
term_years	Length of one term in years: 1 or 3.	Created, from Subscriptions Europe › Duration
price	Fee recorded on the subscription, in euros. 0 = free. A recorded price, not proof of payment.	Original: Subscriptions Europe › Current price
start_date	Date this term began. For records holding several terms, each later term starts where the previous one ended.	Created, from Start date
due_date	Date this term ended: the renewal decision date. Uses the renewal date, or the cancellation date if cancelled earlier.	Created, from Renewal date & Cancelled at

Column	What it means	Source
spell_start	First day of the spell this decision belongs to.	Created
tenure_years	How many years the person had been a member (from spell_start) when this decision came up.	Created
tenure_reliable	False when the spell began with a record from the June 2023 migration. Their real tenure is longer than shown.	Created
next_class	Class of the member's next term (Free, Full or Explorer). Blank if there was no later term. Shows moves such as Free to Full.	Created
next_grade	Grade of the member's next term. Blank if there was no later term.	Created
days_to_next	Days between the due date and the start of the next term. Negative = renewed early, 0 = straight on, blank = never came back.	Created
renewed	True if the next term started within 1 month (31 days) after the due date.	Created
reinstated_later	True if the member did not renew within 1 month but came back later with a new membership.	Created
other_plan	Plan of a subscription in another SETAC region that started within 180 days of the due date. Blank if none.	Original: Subscriptions All › Subscription plan
outcome	Final result of the decision: renewed (2,956), expired (8,294, did not renew within 1 month), moved_unit (22, moved to another SETAC region), pending (61, ended within 1 month of the cut-off, so still in the grace period), not_yet_due (3,837, due after the cut-off).	Created
state	resolved, pending or not_yet_due. Only resolved decisions have a known result.	Created
legacy_product_ended	True for Explorer or 3-year terms ending after 31 August 2026, which could not be renewed as the same product.	Created
in_headline_rate	Simply means should this row count in the analysis for renewal rate. True for the 11,270 rows to include when calculating renewal rates: resolved	Created

Column	What it means	Source
	and not a withdrawn product.	
first_migration_renewal	Simply means was this the first renewal after the system change? True for the first renewal decision of a record carried over in the June 2023 migration (4,242 rows; 2,573 of them due on 1 July 2024). This round renewed far less often than later ones.	Created, from Date created
country	Member's country. "Unknown" if the contact gave none.	Original: Individual Contacts › Default address - Country
geo_unit	SETAC region for that country.	Original: Country list › Geo unit
fee_band	Standard, or Discounted for lower-income countries.	Original: Country list › Income level
sector	Where the member works, such as Academia or Business, industry. "Unknown" if not given.	Original: Individual Contacts › Sector
retired_flag	1 = ticked "Retired or currently not working", 0 = not Retired. Blank = not answered.	Original: Individual Contacts › Retired or currently not working
grad_is_default	True when the graduation date is 1 June 2018 (6224 members), a placeholder from the migration rather than a real date.	Created, from Individual Contacts › Graduation date

Sheet 2: B_year_end_members

One row is one person who was a member on a year-end date. The sheet has 16,361 rows and 14 columns: 5,498 members at 31 December 2023, 3,808 at 31 December 2024, 3,320 at 31 December 2025 and 3,735 at the 30 September 2026 cut-off. A person appears once for each year they were a member. Someone holding two subscriptions on the same date is counted once, at their highest class

(Full, then Explorer, then Free). Most of the fall from 2023 to 2024 is the first renewal round after the migration, due on 1 July 2024, so 2024 onwards is the reliable trend.

Built from

Original file	Columns taken	Joined on
Subscriptions Europe (through the valid_subscriptions sheet)	Contact - ID, Class, Grade, Duration, Subscription plan, Start date, Renewal date	The starting table: everyone whose term covered the snapshot date
Individual Contacts	Default address - Country, Sector, Retired or currently not working	Contact ID
Country list	Geo unit, Income level	Country name

Columns

Column	What it means	Source
contact_id	Anonymised ID of the person.	Original: Subscriptions Europe › Contact - ID
snapshot_year	The year being counted: 2023, 2024, 2025 or 2026.	Created
snapshot_date	The exact date counted: 31 December for 2023 to 2025, and 30 September 2026 (the cut-off) for 2026.	Created
member_class	Class held on that date: Free, Full or Explorer.	Original: Subscriptions Europe › Class
paid_or_free	"Free" for Free members, "Paid" for Full and Explorer.	Created, from Class
grade_short	Grade on that date, without the word "Membership".	Created, from Subscriptions Europe › Grade
member_type	Career group: grade_short, or Retired if the person ticked "Retired or currently not working".	Created, from Grade + Individual Contacts › Retired
plan	Membership plan held on that date.	Original: Subscriptions Europe › Subscription plan
term_years	Length of the term held: 1 or 3 years.	Created, from Subscriptions Europe › Duration

Column	What it means	Source
country	Member's country. "Unknown" if not given.	Original: Individual Contacts › Default address - Country
geo_unit	SETAC region for that country.	Original: Country list › Geo unit
fee_band	Standard or Discounted.	Original: Country list › Income level
sector	Where the member works. "Unknown" if not given.	Original: Individual Contacts › Sector
retired_flag	1 = ticked "Retired or currently not working", 0 = not Retired, blank = not answered.	Original: Individual Contacts › Retired or currently not working

The 2026 figure is at 30 September, not year-end, so it is not a full year.

Sheet 3: C_member_cohorts

One row is one person, described by their first subscription and whether they were still a member 1, 2 and 3 years later. The sheet has 11,141 rows and 25 columns. Real join dates only exist from July 2023 onwards, so earlier members form one Legacy group of 4,441 people. The cohorts are 2023 (1,246), 2024 (2,418), 2025 (1,739) and 2026 (1,290), plus 7 people whose first start is in 2016 or 2022.

Built from

Original file	Columns taken	Joined on
Subscriptions Europe (through the valid_subscriptions sheet)	Contact - ID, Start date, Class, Grade, Date created, Renewal date	The starting table: each person's earliest subscription
Individual Contacts	Default address - Country, Graduation date, Sector, Retired or currently not working, Opted in to be a mentor, Topics interested in	Contact ID
Country list	Geo unit, Income level	Country name

Columns

Column	What it means	Source
contact_id	Anonymised ID of the person. One row each.	Original: Subscriptions Europe › Contact - ID
first_start	Start date of the person's earliest subscription	Created, from Subscriptions Europe ›

Column	What it means	Source
	in the data.	Start date
member_class	Class of that first subscription: Free, Full or Explorer.	Original: Subscriptions Europe › Class
grade	Grade of that first subscription, in full, such as "Student Membership".	Original: Subscriptions Europe › Grade
first_migrated	True if the first record was created before 1 July 2023 (the migration), so the real join date is unknown.	Created, from Subscriptions Europe › Date created
first_start_is_default	True if the first start date is 2 July 2023, a placeholder used for 31.6% of subscriptions.	Created, from Subscriptions Europe › Start date
cohort	Year the person joined, or "Legacy (joined before Jul 2023, date unknown)".	Created
retained_1yr	True if the person was still a member 1 year (plus the 1-month grace period) after first_start.	Created
observable_1yr	True if enough time has passed before the cut-off to check the 1-year result. Leave out rows marked False.	Created
retained_2yr	Same as retained_1yr, at 2 years.	Created
observable_2yr	Same as observable_1yr, at 2 years.	Created
retained_3yr	Same as retained_1yr, at 3 years.	Created
observable_3yr	Same as observable_1yr, at 3 years.	Created
country	Member's country. "Unknown" if not given.	Original: Individual Contacts › Default address - Country
graduation_date	Graduation date the person gave. SETAC uses it to set the grade automatically.	Original: Individual Contacts › Graduation date
sector	Where the member works. "Unknown" if not given.	Original: Individual Contacts › Sector
retired_flag	1 = ticked "Retired or currently not working", 0 = not Retired, blank = not answered.	Original: Individual Contacts › Retired or

Column	What it means	Source
		currently not working
mentor	1 if the person volunteered as a mentor, 0 if not.	Original: Individual Contacts › Opted in to be a mentor
grad_is_default	1 when the graduation date is the 1 June 2018 placeholder.	Created, from Graduation date
n_topics	Number of science topics the person said they are interested in. A preference, not an activity.	Created, from Individual Contacts › Topics interested in
geo_unit	SETAC region for that country.	Original: Country list › Geo unit
fee_band	Standard or Discounted.	Original: Country list › Income level
grade_short	Grade without the word "Membership".	Created, from Grade
member_type	Career group: grade_short, or Retired.	Created, from Grade + Retired
paid_or_free	"Free" or "Paid", from the first subscription's class.	Created, from Class

Sheet 4: person_details

One row is one contact in SETAC's system, member or not. The sheet has 28,922 rows and 10 columns. It is Individual Contacts joined to the Country list, and it supplies the personal details added to Sheets A, B and C. Members should not be counted from this sheet as it also holds staff, suppliers and non-members.

Built from

Original file	Columns taken	Joined on
Individual Contacts	Contact ID, Default address - Country, Graduation date, Sector, Retired or currently not working, Opted in to be a mentor, Topics interested in	The starting table
Country list	Geo unit, Income level	Country name. All country names matched.

The 26 columns dropped from Individual Contacts are personal details not needed (job title, organisation, pronouns), the 11 directory display settings (67% blank), Loyalty badge (94% blank, with two conflicting definitions), spare IDs, email permission flags, and system dates.

Columns

Column	What it means	Source
contact_id	Anonymised ID of the contact. One row each.	Original: Individual Contacts › Contact ID
country	Country of the contact's address. Set to "Unknown" for the 1,995 contacts who gave none.	Original: Individual Contacts › Default address - Country
graduation_date	Graduation date given (past or expected). 512 are blank.	Original: Individual Contacts › Graduation date
sector	Where the contact works. Set to "Unknown" when blank.	Original: Individual Contacts › Sector
retired_flag	1 = ticked "Retired or currently not working", 0 = not Retired, blank = not answered (910).	Original: Individual Contacts › Retired or currently not working
mentor	True if the contact volunteered as a mentor.	Original: Individual Contacts › Opted in to be a mentor
grad_is_default	True when the graduation date is the 1 June 2018 placeholder (6,224 contacts).	Created, from Graduation date
n_topics	Number of science topics the contact chose. 0 = none chosen.	Created, from Individual Contacts › Topics interested in
geo_unit	SETAC region for that country. "Unknown" when the country is unknown.	Original: Country list › Geo unit
fee_band	Standard or Discounted. "Unknown" when the country is unknown.	Original: Country list › Income level

Sheet 5: valid_subscriptions

One row is one membership term that passed the cleaning rules. The sheet has 16,046 rows, 31 columns and 11,141 people. It started as 16,234 subscription records; 2,740 were removed (see exclusion_counts), leaving 13,494, which were then split into 16,046 terms. Sheets A, B and C are all built from it.

Built from

Original file	Columns taken	Joined on

Original file	Columns taken	Joined on
Subscriptions Europe	Contact - ID, ID, Contact - Contact type, Contact - Contact status, Status, Class, Grade, Duration, Subscription plan, Start date, Renewal date, Cancelled at, Sub status, Current price, Date created (15 of 47 columns)	No join: one file only

The 32 columns dropped were empty (such as Pause from and Renewal status), almost always blank renewal-switch and payment-process fields, repeats of other columns (Package, Product/Package, Current grade, Renewal grade), or system sync dates.

Columns

Column	What it means	Source
contact_id	Anonymised ID of the person.	Original: Subscriptions Europe › Contact - ID
subscription_id	Unique ID for this term: the record ID plus "_t" and the term number.	Created, from Subscriptions Europe › ID
contact_type	Type of contact. Always "Contact" here, because staff, sales and other non-members were removed.	Original: Subscriptions Europe › Contact - Contact type
contact_status	Status of the contact record: Active or Inactive. Not membership status. Merged duplicate profiles have been removed.	Original: Subscriptions Europe › Contact - Contact status
status	SETAC's status for the whole record: Active, Expired, Cancelled, Lapsed or Pending. Not used to decide renewal.	Original: Subscriptions Europe › Status
member_class	Free, Full or Explorer.	Original: Subscriptions Europe › Class
cancelled_at	Date and time the record was cancelled. Blank if never cancelled.	Original: Subscriptions Europe › Cancelled at
price	Recorded fee in euros.	Original: Subscriptions Europe › Current price

Column	What it means	Source
duration	"1 year" or "3 years".	Original: Subscriptions Europe › Duration
plan	Membership plan.	Original: Subscriptions Europe › Subscription plan
grade	Grade in full, such as "Professional Membership".	Original: Subscriptions Europe › Grade
sub_status	"Grade changed" marks a record replaced because the grade changed, which is not a renewal. "Renewable" or blank otherwise.	Original: Subscriptions Europe › Sub status
renewal_date	SETAC's renewal date for the whole record: the end of its latest term.	Original: Subscriptions Europe › Renewal date
start_date	Date this term began. The first term uses the record's start date; each later term starts where the previous one ended.	Created, from Start date
date_created	When SETAC's system created the record. 26 June 2023 marks the migration.	Original: Subscriptions Europe › Date created
term_years	Length of one term: 1 or 3.	Created, from Duration
term_end	Record start date plus one term. Kept as a check against renewal_date.	Created
end_date	Date this term ended. The last term ends on the renewal date, or the cancellation date if earlier.	Created
start_is_default	True if the record's start date is 2 July 2023, a migration placeholder.	Created
migrated_record	True if the record was created before 1 July 2023.	Created, from Date created
exclude_reason	Always "KEPT" here, as only kept rows are in this sheet.	Created
n_terms	Number of terms the original record covered.	Created

Column	What it means	Source
	11,881 records hold 1 term.	
record_id	Original subscription ID, before splitting into terms.	Original: Subscriptions Europe › ID
term_no	Position of this term within its record: 1, 2, 3 and so on.	Created
running_max_end	Working column: latest end date the person had reached up to this term.	Created
previous_max_end	Working column: latest end date before this term. Blank for the person's first term.	Created
extends_cover	True if this term adds new membership time rather than overlapping time already covered. Only these terms create renewal decisions.	Created
new_spell	True if this term starts a new spell (first term, or a return after more than 1 month away).	Created
spell_no	Spell number: 1, 2 or 3.	Created
spell_start	First day of the spell.	Created
spell_first_migrated	True if the spell began with a migrated record, so tenure is undercounted.	Created

Sheet 6: excluded_subs_blank_co_id

One row is one subscription record removed from the analysis because it has no Contact ID. The sheet has 2,238 rows and 20 columns, 13.8% of all records in Subscriptions Europe. It is kept so we can show what was removed and check whether it biases the results. None of these rows are used in Sheets A, B or C.

The removed records are not a random sample. 96.6% are Free memberships (2,162), against 76.4% of the records kept. 95% started in 2023 (2,126), and 2,070 were created in June 2023, so almost all of them came from the system migration. None is still active: 2,154 are Expired and 84 Cancelled.

Built from

Original file	Columns taken	Joined on
Subscriptions Europe	Contact - ID, ID, Contact - Contact type, Contact - Contact status, Status, Class, Grade, Duration, Subscription plan, Start date, Renewal date, Cancelled at, Sub status, Current price, Date	No join: without a Contact ID, these rows cannot be linked to any other

Original file	Columns taken	Joined on
	created	file

These are the same 15 columns as valid_subscriptions, plus 5 date columns the notebook created before the rows were set aside. The records were not split into terms, so each row is a whole record.

Columns

Column	What it means	Source
contact_id	Always blank: this is why the row was excluded.	Original: Subscriptions Europe › Contact - ID
subscription_id	SETAC's ID for the subscription record. Unlike valid_subscriptions, it has no "_t" ending, because the record was not split into terms.	Original: Subscriptions Europe › ID
contact_type	Always blank. It comes from the linked contact, and there is none.	Original: Subscriptions Europe › Contact - Contact type
contact_status	Always blank, for the same reason.	Original: Subscriptions Europe › Contact - Contact status
status	SETAC's status for the record: Expired (2,154) or Cancelled (84).	Original: Subscriptions Europe › Status
member_class	Free (2,162), Full (70) or Explorer (6).	Original: Subscriptions Europe › Class
cancelled_at	Date and time the record was cancelled. Blank for the 2,154 never cancelled.	Original: Subscriptions Europe › Cancelled at
price	Recorded fee in euros. 0 = free.	Original: Subscriptions Europe › Current price
duration	"1 year" (2,229) or "3 years" (9).	Original: Subscriptions Europe › Duration
plan	Membership plan: annual or 3-year, standard or discounted.	Original: Subscriptions Europe › Subscription plan
grade	Grade in full, such as "Professional Membership". 4 rows say "Staff", which suggests SETAC staff records.	Original: Subscriptions Europe › Grade
sub_status	"Grade changed" (96 rows) marks a record replaced because the grade changed. Blank	Original: Subscriptions Europe › Sub status

Column	What it means	Source
	otherwise.	
renewal_date	SETAC's renewal date for the record: when its membership ended or was due.	Original: Subscriptions Europe › Renewal date
start_date	Date the record started. 2 July 2023 is the migration placeholder.	Original: Subscriptions Europe › Start date
date_created	When SETAC's system created the record. 26 June 2023 marks the migration.	Original: Subscriptions Europe › Date created
term_years	Length of one term: 1 or 3.	Created, from Duration
term_end	Start date plus one term. A check against renewal_date.	Created
end_date	Date the record's membership ended: the renewal date, or the cancellation date if earlier.	Created
start_is_default	True if the start date is the 2 July 2023 placeholder.	Created, from Start date
migrated_record	True if the record was created before 1 July 2023.	Created, from Date created

Summary sheets: exclusion_counts and year_end_comparison

Both sheets are counts, not data about individual people. Neither joins any files: they summarise Subscriptions Europe after the cleaning steps.

exclusion_counts

One row is one reason a subscription record was kept or removed. The 7 rows add up to all 16,234 records in Subscriptions Europe. The reasons are applied in the order listed, and each record gets the first reason that fits.

Column	What it means	Source
reason	Why the record was kept or removed (see below).	Created
rows	Number of subscription records with that reason.	Created

reason value	Meaning	Records
KEPT	Valid membership record, used in the analysis	13,494
no_contact_id	No Contact ID, so it cannot be linked to a person.	2,238

reason value	Meaning	Records
	Listed in excluded_subs_blank_co_id	
staff	Belongs to a SETAC staff member. SETAC asked for staff to be excluded	22
not_a_member_contact_type	Belongs to sales, press, accountants or another non-member contact	34
merged_duplicate_contact	Belongs to a duplicate profile SETAC merged into another profile of the same person. The person carries on under another ID that cannot be linked, so keeping these would count them as lost	409
deceased	Belongs to a contact marked as deceased	8
never_valid_or_zero_length	Ended on or before its start date, for example cancelled before it began	29

year_end_comparison

One row is one snapshot year. It shows how much the year-end member count drops.

Column	What it means	Source
index	The snapshot year (2023 to 2026).	Created
linked_members	Real member count, the figure used in Sheet B. It counts people, each once, after all the cleaning (no blank IDs, no staff, no merged duplicates, no deceased).	Created
all_valid_subscription_rows	It counts subscription records from the original Subscriptions Europe file that were running on that date, with almost nothing removed.	Created
difference	all_valid_subscription_rows minus linked_members.	Created

Year	linked_members	all_valid_subscription_rows	difference
2023	5,498	7,821	2,323
2024	3,808	3,966	158
2025	3,320	3,381	61
2026	3,735	3,752	17